ELLab

SUNGKYUNKWAN UNIVERSITY

Anomaly Segmentation with SAM

2026.04.28

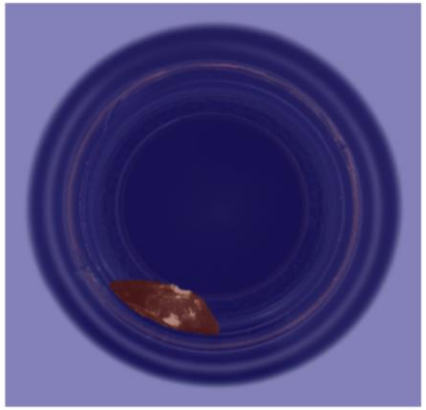
Experiment – A.S with SAM3

Baseline A.S.

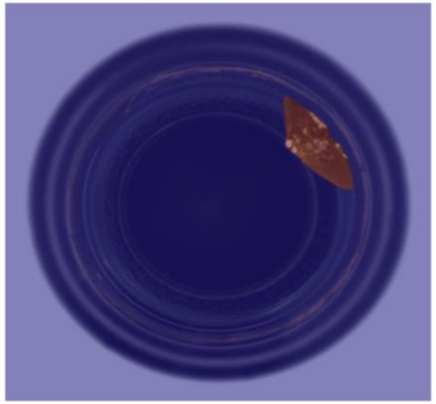
Input: bottle test image & specie name text prompt

- 63개의 이미지 중 3개의 이미지에서만 anomaly segmentation 수행됨
- threshold = 0.5 (기본 설정)
- 세개의 이미지 모두 defect type broken small임
  - ➢ Defect type에 따른 segmentation 차이 존재 할 것임

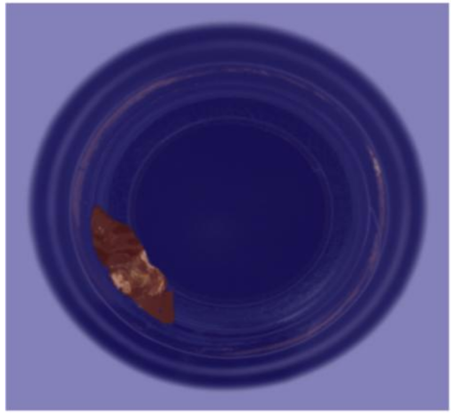
0 | score=0.859



0 | score=0.766



4 | score=0.594

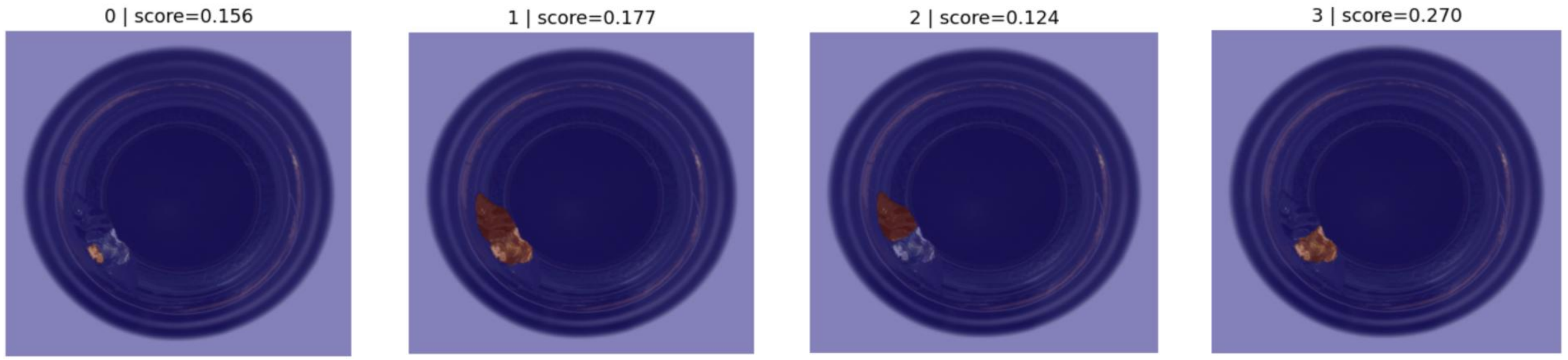


Experiment – A.S with SAM3

Threshold adjusting

Input: bottle test image & specie name text prompt

- threshold = 0.1 (기본 설정)
- 63개의 이미지 중에서 14개 이미지 segmentation 됨
- 한 이미지에 대해 최소 mask 최소 1개 최대 8개까지 나옴
- mask score와 실제 segmentation 품질 간의 상관관계는 없어 보임
- 14개의 이미지 defect type 모두 broken이고 contamination type은 한번도 segmentation 되지 않음

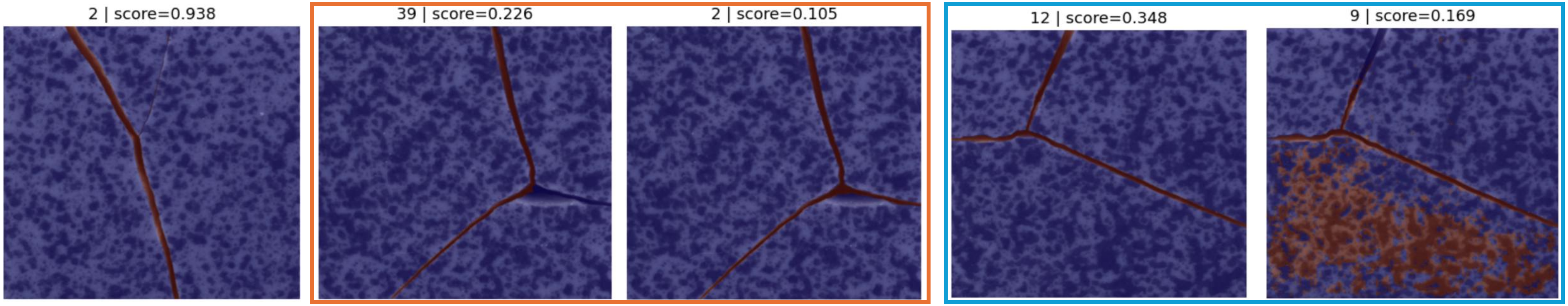


Experiment – A.S with SAM3

Another dataset

Input: tile test image & specie name text prompt

- threshold = 0.1 (기본 설정)
- 84개의 abnormal이미지 중에서 50개 이미지 segmentation 됨
- 한 이미지에 대해 최소 mask 최소 1개 최대 40개까지 나옴



Experiment – A.S with SAM3

Conclusion

- Abnormal 이미지를 segmentation 하는 비율 높여 야함
  - Img class, Defect type별 편차 해결
  - Text 이외의 prompt 제공

- 출력되는 여러 mask들 중에 score 이외의 판별 기준 필요
  - Reinforce learning, GRPO (한 이미지에 대해 여러 output 나오므로 상대 비교 가능하도록 reward 설계)
  - 별도의 학습 network 만들어서 mask 선택 단계에 붙이기

"Abnormal 이미지 segmentation 비율 높이고 mask 선택 잘 하자!"

Procedure

- A.D using foundation model papers
  - WinCLIP
  - AnomalyCLIP

- SAM1,2,3 papers
  - Experiment with mvTec

→ Zero-Shot Anomaly Segmentation (ZSAS)

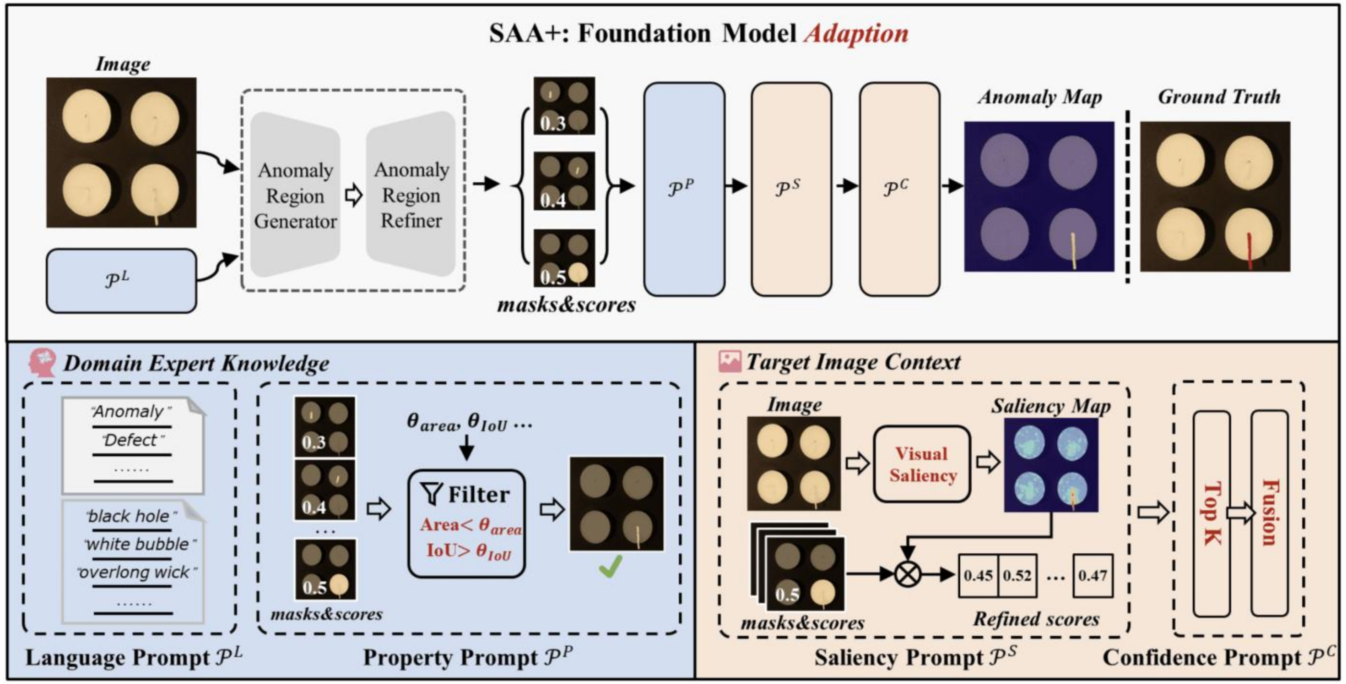
- ZSAS with SAM
  - SAA, SAA+
  - ClipSAM

- ZSAS without SAM
  - MuSc → mask prompt

- Mask Selection
  - IAD-R1 Paper → RL method (GRPO)

Anomaly Region Generator

- Language Prompt
  - “anomaly”, “defect” + class-specific description
- Text prompt + Image → Grounding DINO
- anomaly candidate boxes 생성



SAM Refinement

- Input: Image I, Generated boxes R
- SAM (prompt-based segmentation)
- Pixel-level masks 생성

SAA+

Property prompt (Rule-based Filtering)

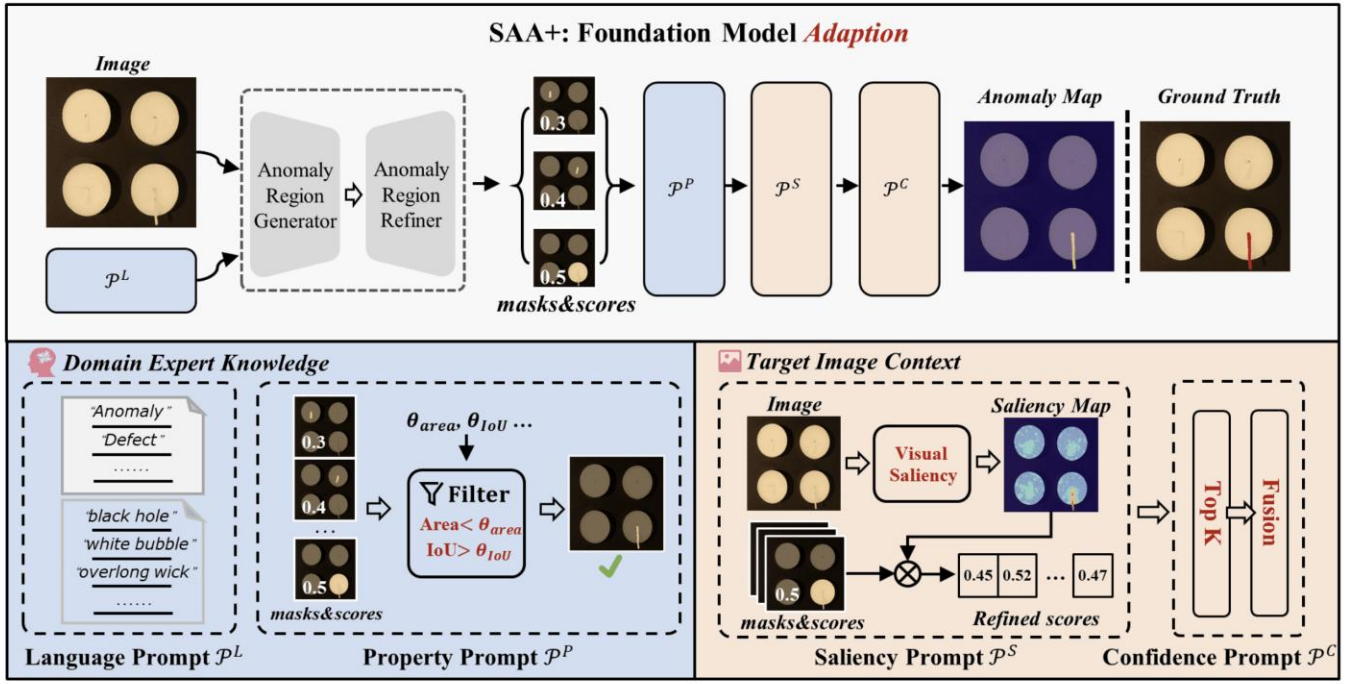
- IoU threshold를 통해 노이즈 제거
- area 기준으로 anomaly 크기 제한

Saliency prompt

- feature similarity 기반 saliency map 계산
- region 단위 saliency score 산출

Confidence Prompt (Top-K Selection)

- score 기준 상위 K개 선택



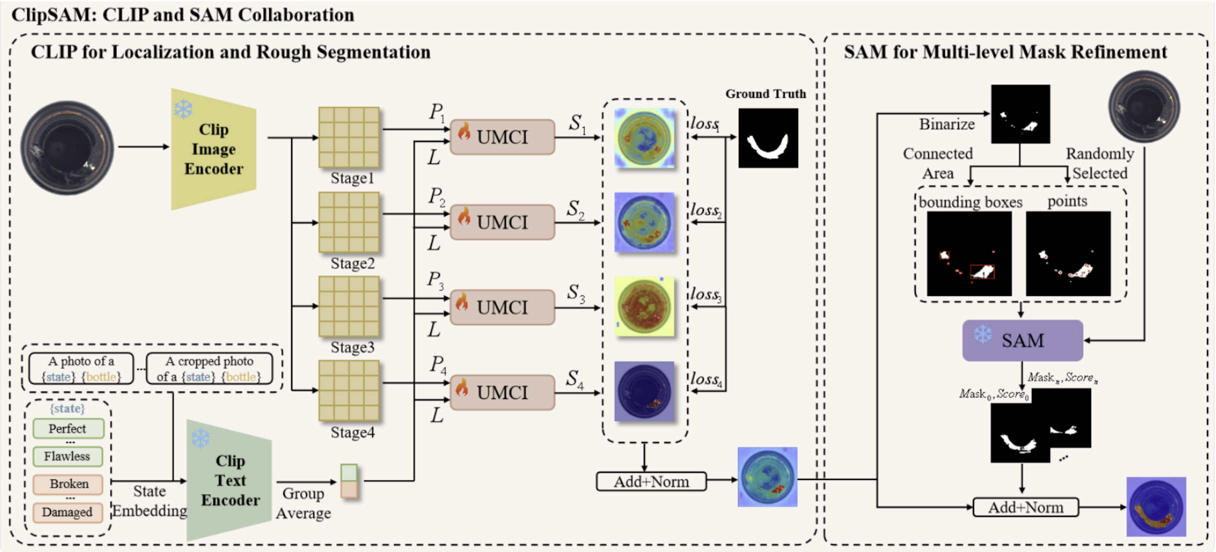
Final Aggregation

여러 region mask를 confidence 기반으로 weighted sum

ClipSAM CLIP and SAM Collaboration for Zero-Shot Anomaly Segmentation (arXiv)

CLIP Feature Extraction

- Image encoder + Text encoder
- Text prompt는 여러 Group average 통해서 prototype vector 만들기 (like WinCLIP)



Similarity Map 생성

- image feature vs text feature 비교
- pixel/patch 단위 similarity

Prompt Generation

similarity map → thresholding

anomaly 높은 영역 선택하여 points / bounding boxes 생성

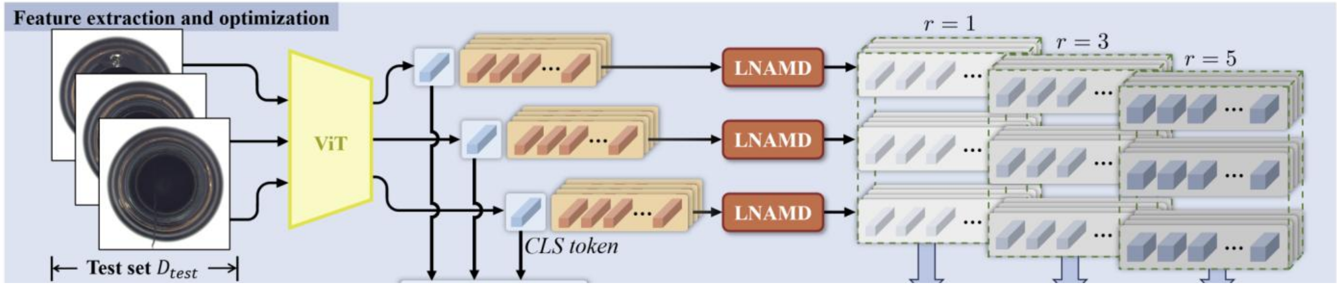
SAM Refinement prompt 받아서 pixel level segmentation 수행

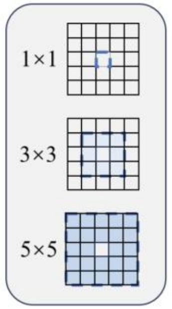
MUSC (ICLR 2024)

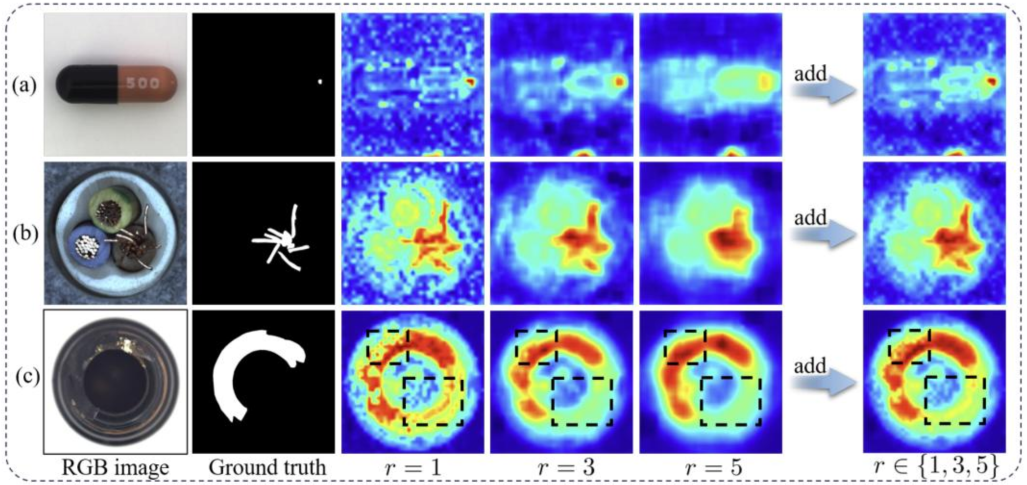
Local Neighborhood Aggregation (LNAMD)

각 patch feature를 다양한 크기의 neighborhood 기반으로 재구성 → multi-scale representation 생성

- Input: ViT patch feature $F_i \in \mathbb{R}^{M \times C}$
- Local Aggregation
  - 각 patch에 대해 주변 $r \times r$영역 평균 pooling
  - $r \in \{1, 3, 5\}$
- Multi-scale Feature
  - $r = 1$: local detail (작은 anomaly 대응)
  - $r = 3$: 중간 수준 context
  - $r = 5$: 넓은 영역 (큰 anomaly 대응)
- Stage-wise 적용
  - ViT 여러 stage에서 LNAMD 수행
  - 서로 다른 수준의 feature (low-level ~ high-level) 활용





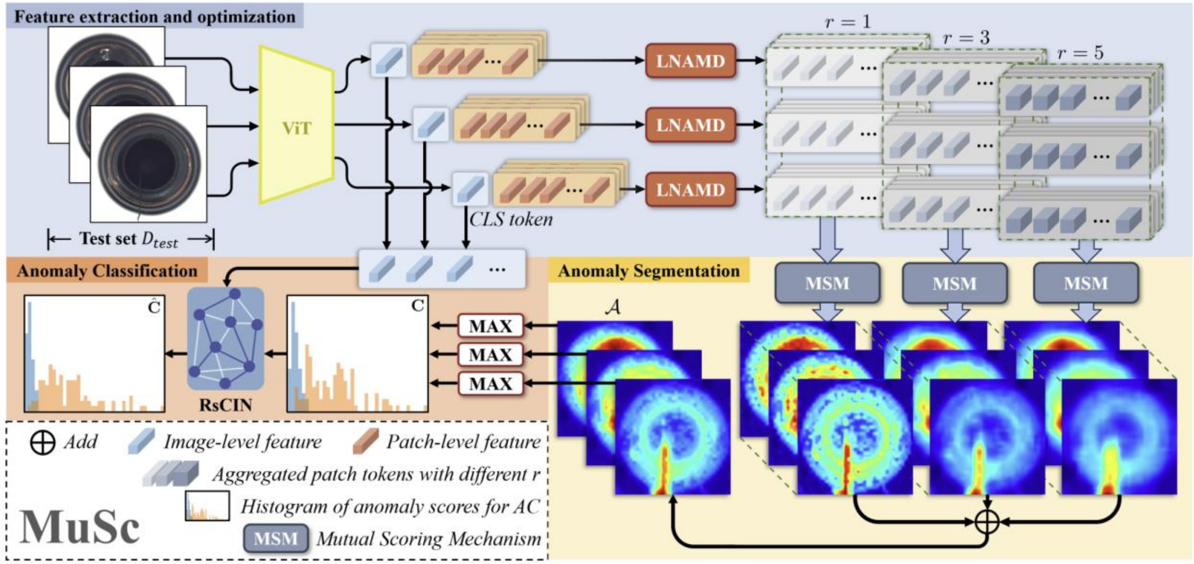


MUSC

Mutual Scoring Mechanism (MSM)

unlabeled test images 간 비교를 통해 patch-level anomaly score 계산

- Input
  - LNAMD로 생성된 multi-scale patch feature
  - test image set D_u



Patch-level Scoring

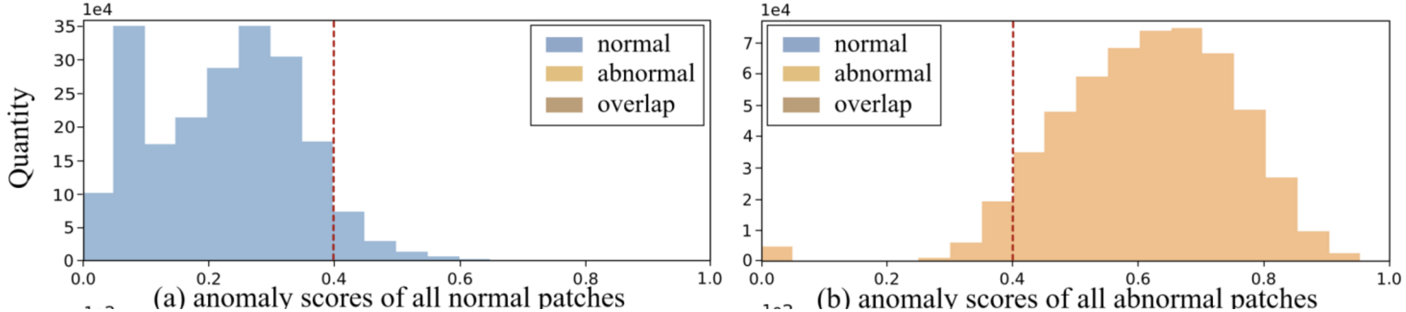
각 patch를 다른 이미지들과 비교하여 anomaly score 산출

- 현재 patch ↔ 다른 이미지의 모든 patch 비교

$$a_{i,l}^{m,r}(I_j) = \min_n \|\hat{p}_{i,l}^{m,r} - \hat{p}_{j,l}^{n,r}\|_2$$

$$A_{i,l}^{m,r} = [a(I_1), a(I_2), \ldots, a(I_{N-1})]$$

MUSC



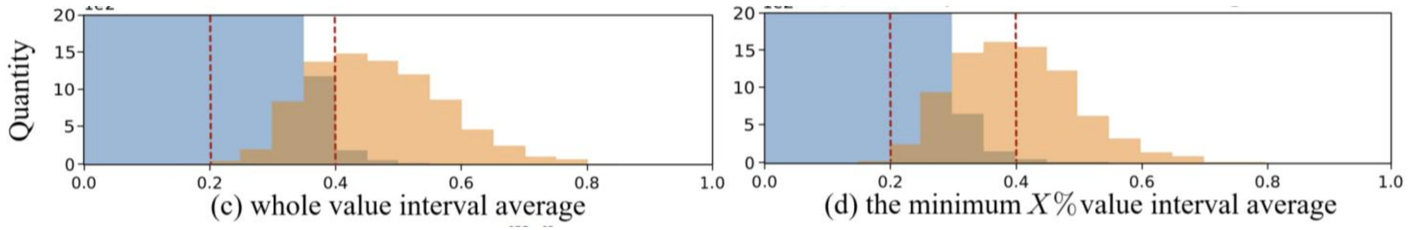
Interval Average (IA)

노이즈 제거를 위해 일부 score만 사용

- X% 구간만 평균 $\bar{a}_{i,l}^{m,r} = \frac{1}{K}\sum_{k=1}^{K} a_{i,l}^{m,r}(\overline{I_k})$
- 정상 patch: 대부분 이미지에서 매우 유사 → 낮은 score 유지
- 이상 patch: 전반적으로 유사 patch 없음 → 높은 score 유지

Multi-scale Aggregation

- 여러 stage + 여러 r결과 평균
- $a_i^m = \frac{1}{L}\sum_{l=1}^{L}\frac{1}{3}\sum_{r\in\{1,3,5\}}\bar{a}_{\mathrm{i,l}}^{\mathrm{m,r}}$



Output

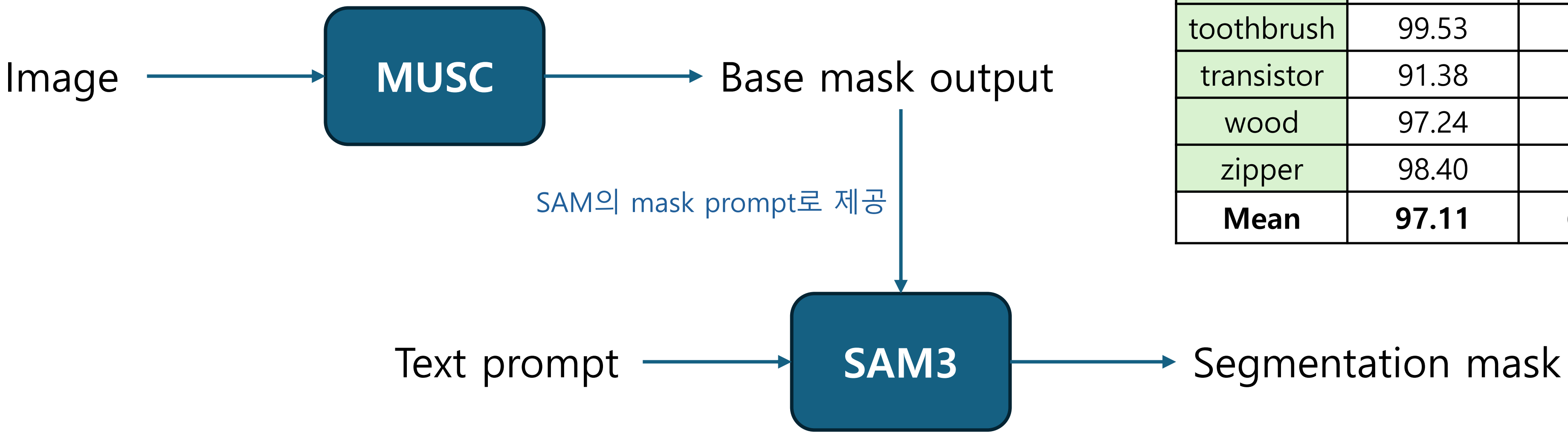
- Patch-level anomaly map (segmentation) $A_i = [a_i^1, \dots, a_i^M]^{\top}$
- Image-level score = max patch score $c_i = \max(a_i^1, \dots, a_i^M)$

Experiment

Mask Prompt Extraction with MUSC

Plan A

- MUSC 사용해서 base mask 추출함
  - 모든 object에 대해서 anomaly score 계산함
  - heatmap png, npy 파일로 저장함
- 추출된 mask를 SAM의 프롬프트로 주고 더 정교한 Segmentation 결과를 얻음



Class	AUROC	F1	AP	AUPRO
bottle	98.48	79.17	83.04	96.10
cable	95.76	60.98	57.71	89.62
capsule	98.96	49.80	48.50	95.58
carpet	99.45	73.33	76.05	97.58
grid	98.16	43.93	38.22	93.89
hazelnut	99.38	73.41	73.28	92.23
leather	99.72	62.85	64.48	98.74
metal_nut	86.12	46.22	47.54	89.34
pill	97.47	65.54	67.25	98.01
screw	98.77	41.87	36.11	94.40
tile	97.90	74.71	78.90	94.65
toothbrush	99.53	70.20	67.79	95.50
transistor	91.38	59.24	58.41	77.23
wood	97.24	68.64	74.74	94.50
zipper	98.40	62.49	61.89	94.83
Mean	**97.11**	**62.16**	**62.26**	**93.48**

Experiment

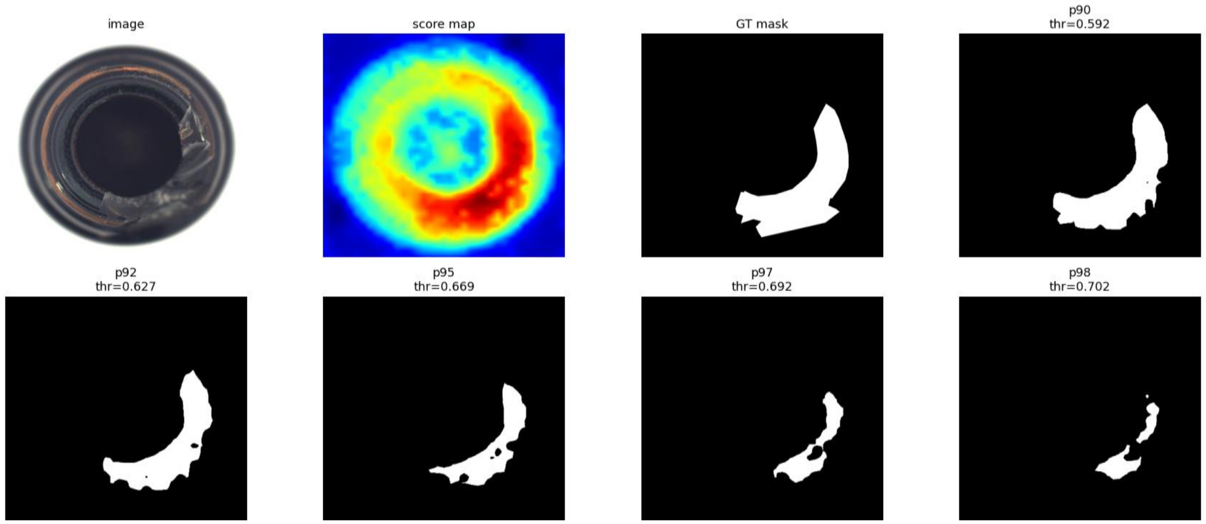
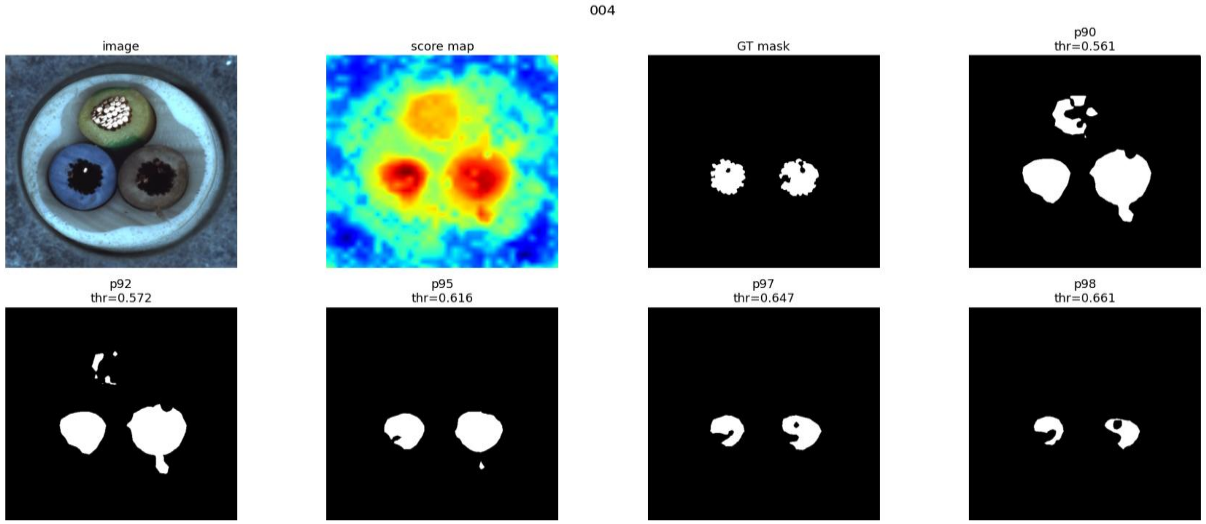


image
score map
GT mask
p90
thr=0.592
p92
thr=0.627
p95
thr=0.669
p97
thr=0.692
p98
thr=0.702


004
image
score map
GT mask
p90
thr=0.561
p92
thr=0.572
p95
thr=0.616
p97
thr=0.647
p98
thr=0.661

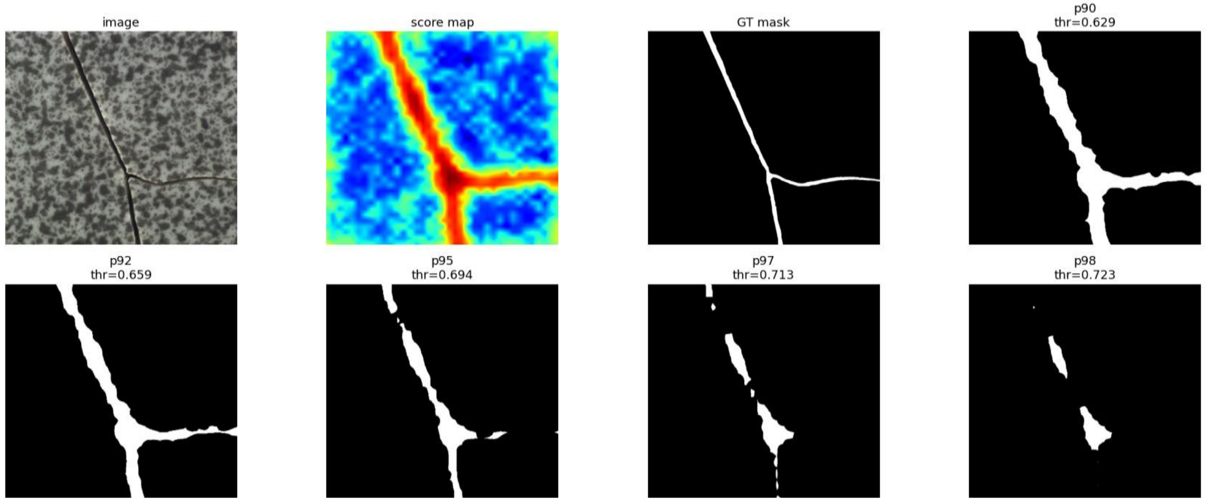


image
score map
GT mask
p90
thr=0.629
p92
thr=0.659
p95
thr=0.694
p97
thr=0.713
p98
thr=0.723

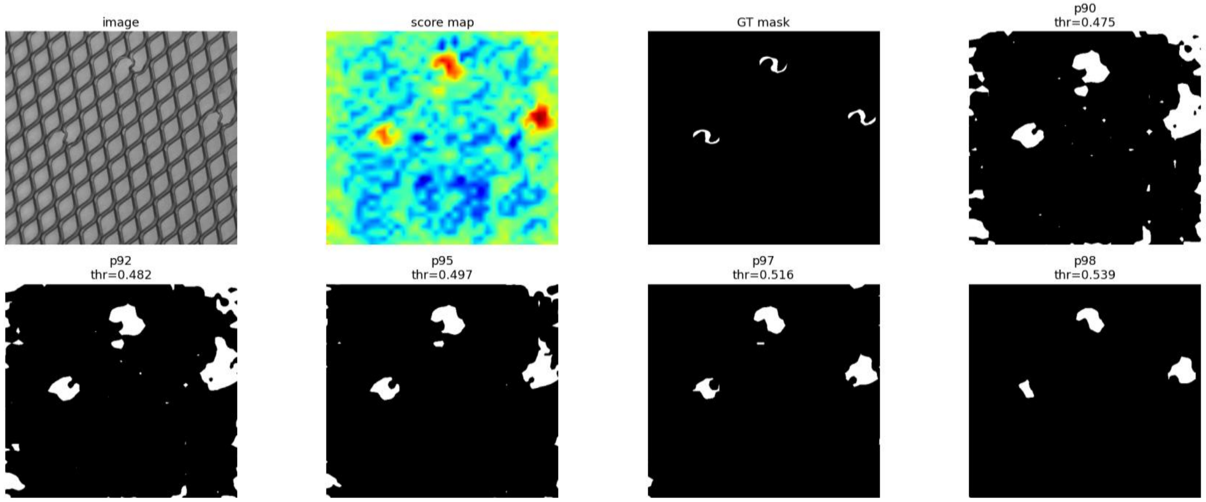


image
score map
GT mask
p90
thr=0.475
p92
thr=0.482
p95
thr=0.497
p97
thr=0.516
p98
thr=0.539

Experiment

Mask Prompt Extraction with MUSC

- Heatmap → binary mask 변환해야함
- Object별 & defect type별 최적의 threshold값
  - 보통 anomaly region이 작으면 threshold 작은게 좋고 넓으면 큰게 좋음
- Threshold 0.92로 잡고 sam3 image processor에 넣으려고 시도
- Mask prompt 받는 API 없어서 구현 실패함
  - Github 코드 찾아보면 분명 mask prompt를 dense embedding으로 입력하는 모듈 있는데 sam3 processor에서는 geometric_prompt로 box만 있음
- 우선 SAM1으로 segmentation 수행함
  - Binary mask 생성까지는 잘 되는데 sam1 모델에 넣으니까 이상해짐
  - Mask prompt 받는 sam1 성능 자체가 안 좋거나 mask가 너무 sparse 할 가능성

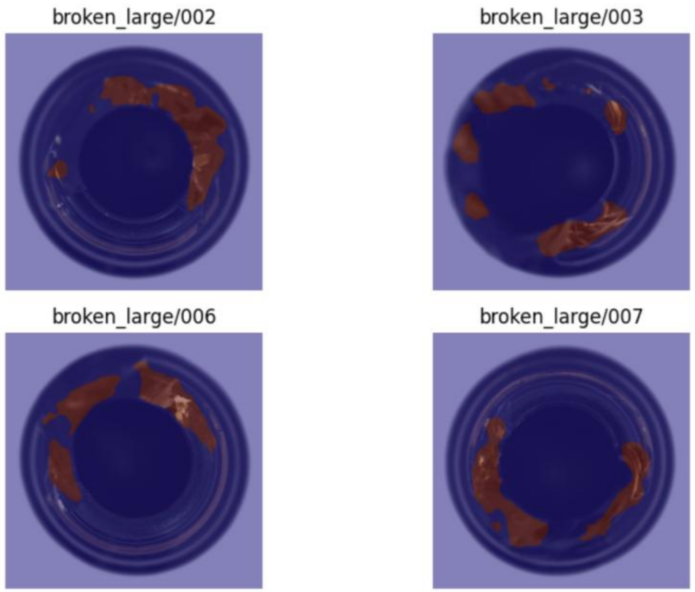
Experiment

Mask Prompt Apply to SAM1

- Threshold 기준으로 binary mask 만들면 defect area 큰 경우 noise와 실제 anomaly 부분 잘 구분 못함

→ 오른쪽 사진처럼 anomaly part가 조각남

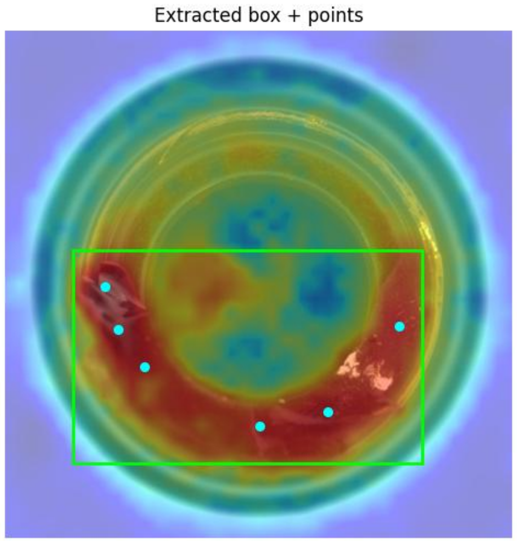
- Binary mask 말고 logit mask 사용해서 연속적인 soft prior를 prompt로 제공
    - 아래 사진처럼 결과를 보면 3개의 mask 후보가 나왔음
    - Mask score와 실제 segmentation 품질 간의 상관관계 낮음
    - Segmentation 품질도 좋지 않음

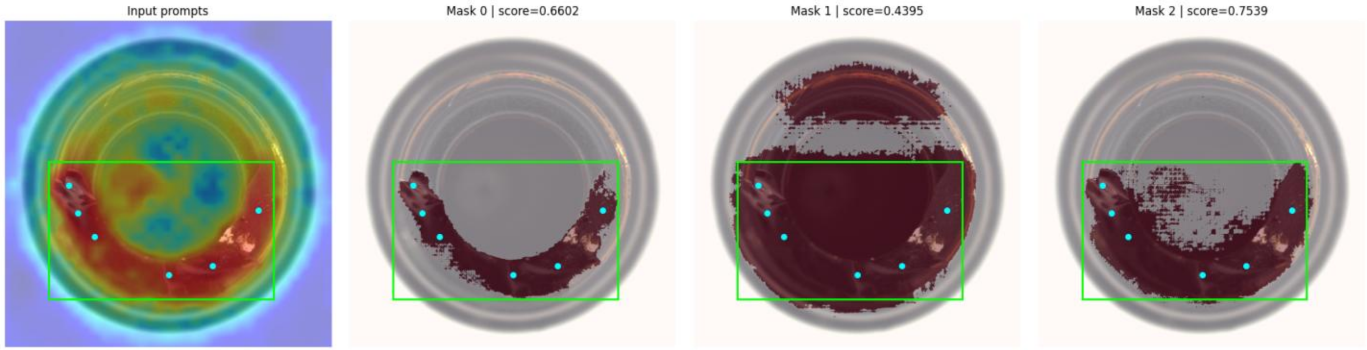


Experiment

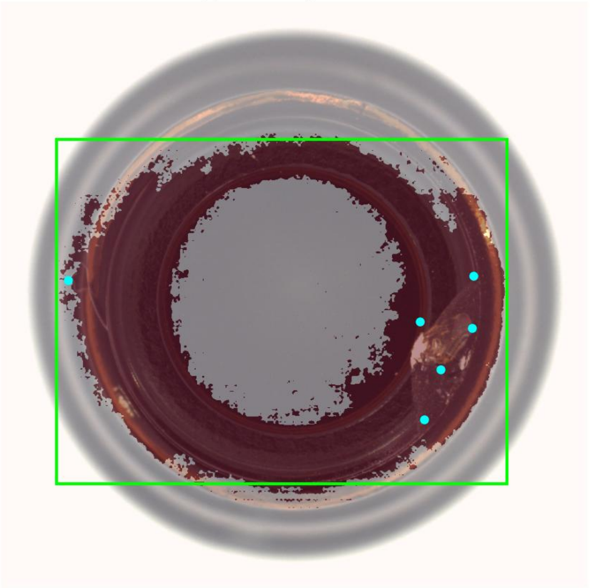
point + box + mask prompt

- Box extraction from heatmap
  - Normalize → blur로 smoothing → percentile threshold로 anomaly 후보 영역 추출 → morphology close로 끊긴 영역 연결 → connected component 분석 → 전체 anomaly 후보를 감싸는 box 생성
- Point extraction from heatmap
  - box 내부 binary 영역에서 서로 멀리 떨어진 point 선택 → point가 부족하면 거리 조건 완화
- Logit mask extraction from heatmap





Experiment



Problem

- Broken part 떨어져 있는 두 부분인 경우 box가 normal인 부분을 많이 포함하게 되므로 noise 정보 제공
- 전체적으로 segmentation 품질 좋지 않음
  - 오히려 text prompt만 사용했을때가 더 나음
- Sam3에서 text prompt랑 mask, point prompt 같이 사용할 수 있는 방법 모르겠음

PLAN

- Sam3에 mask prompt 적용시킬 방법 찾기
- 안되면 box prompt or pointer로 변환해서 적용
- Mask 후보들 중 품질 좋은 mask 선택할 방법 찾기
- Sam3의 text prompt 능력 최대한 활용해보기
  - Sam3 쓰는 이유가 text prompt의 concept 능력 쓰고자 함이기 때문
  - Defect concept + additional prompt

IAD-R1 (AAAI 2026)

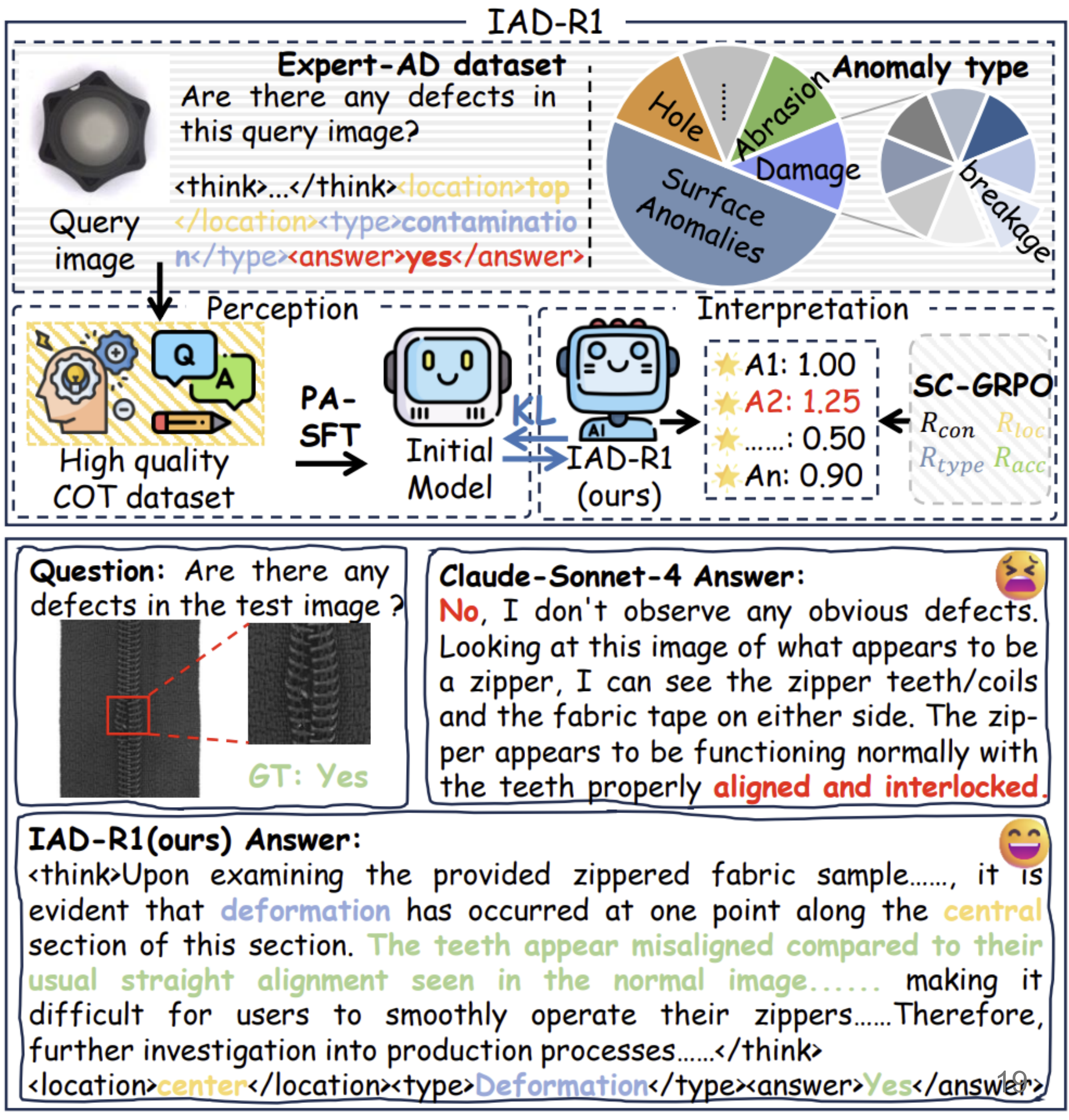
Yes/No → Explainable Anomaly Detection

Two-Stage Framework

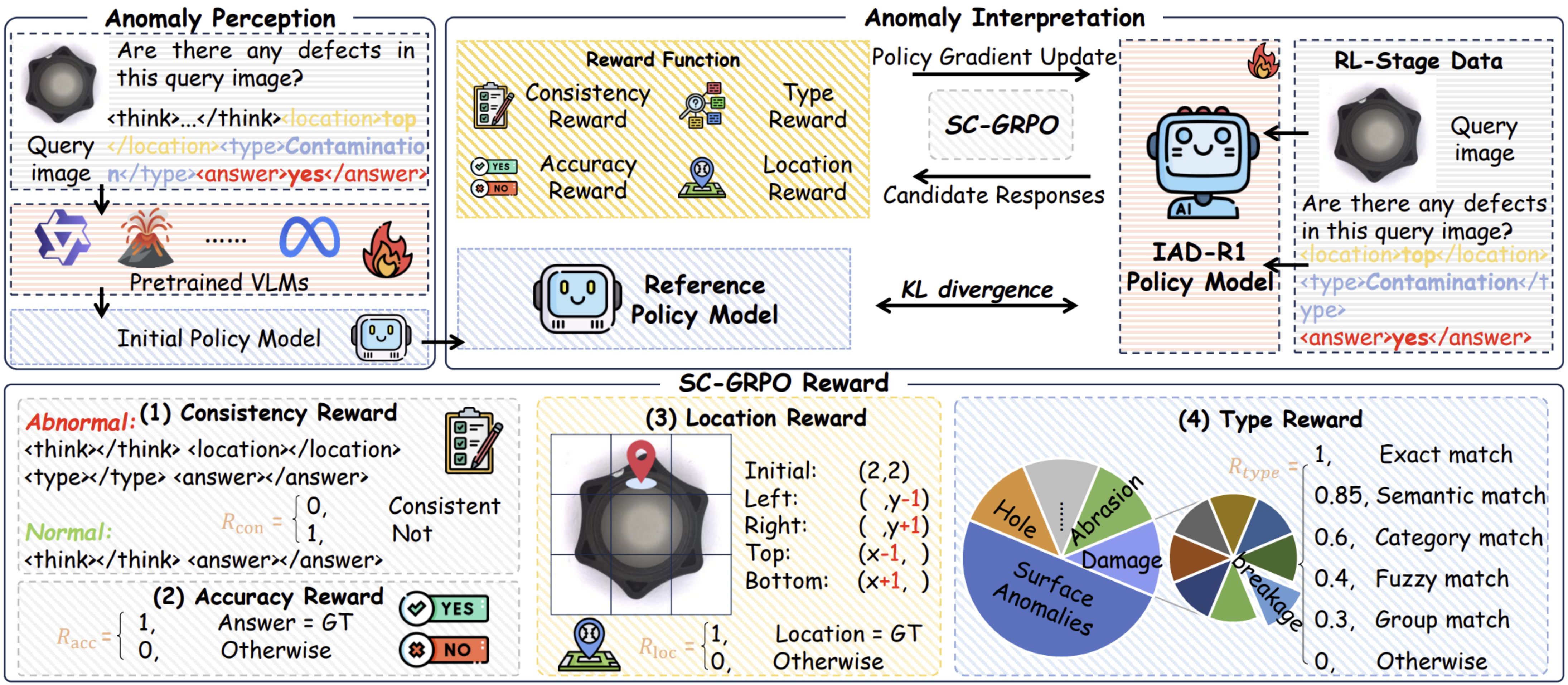
- Stage 1: PA-SFT (Supervised)
  - Expert-AD dataset (with CoT)
  - anomaly perception
- Stage 2: SC-GRPO (Reinforcement Learning)
  - Multi-dimensional reward: Accuracy, Location, Type, Consistency
  - 단순 정답이 아니라 논리적으로 맞는 reasoning 학습

Example

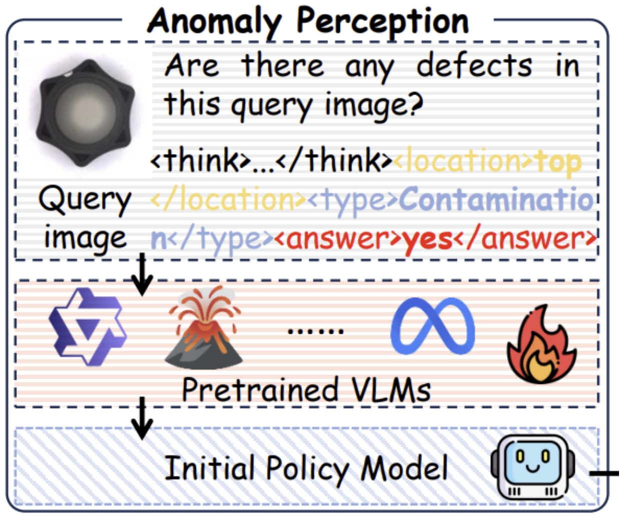
- 기존 모델: 오답 + 잘못된 reasoning
- IAD-R1: 정답 + 위치 + 유형 + 일관된 reasoning



Methodology



Methodology



PA-SFT (Perception Activation SFT)

Training Data

- Dataset: Expert-AD
- Input format: (Image I, Prompt p, Output O)

Output Structure

- Normal image: (Reasoning T + Answer A)
- Anomalous image: (Reasoning T + Location L + Type t + Answer A)

$$\mathcal{L}_{\text{PA-SFT}} = -\mathbb{E}_{(I,p,O)\sim\mathcal{D}_{\text{Expert-AD}}} \sum_{i=1}^{L} \log \pi_\theta(o_i | I, p, o_{<i})$$

Key Idea

- CoT 기반 학습 → structured reasoning 생성 가능 & reasoning ↔ answer 일관성 확보

Training Objective

- 다음 토큰 예측 (Language Modeling) & log-likelihood maximize

Methodology

SC-GRPO (Reinforcement Learning Stage)

Input Model

- Initial policy: PA-SFT trained model

PPO

- single reward + value network

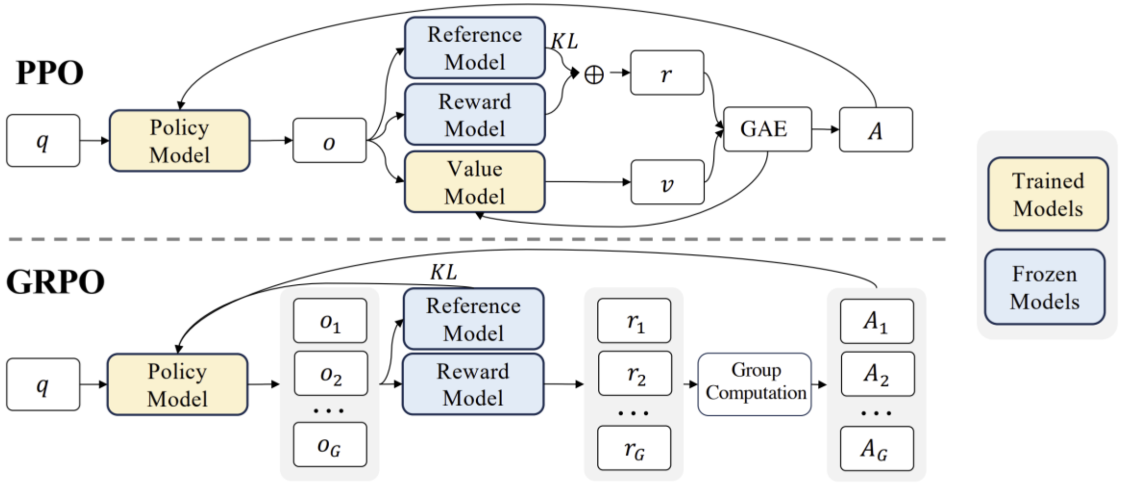
GRPO

- relative reward + no value network

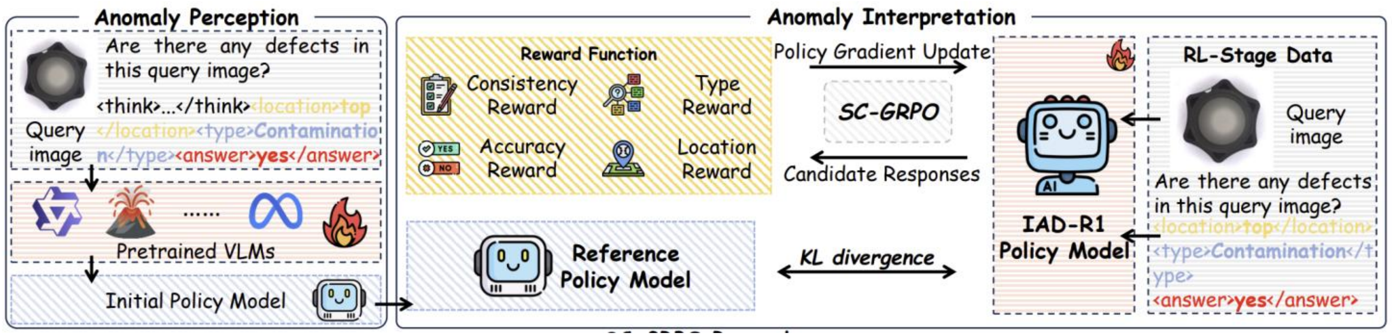
SC-GRPO

- relative + multi-reward + structured reasoning

Many outputs from one query → select GRPO



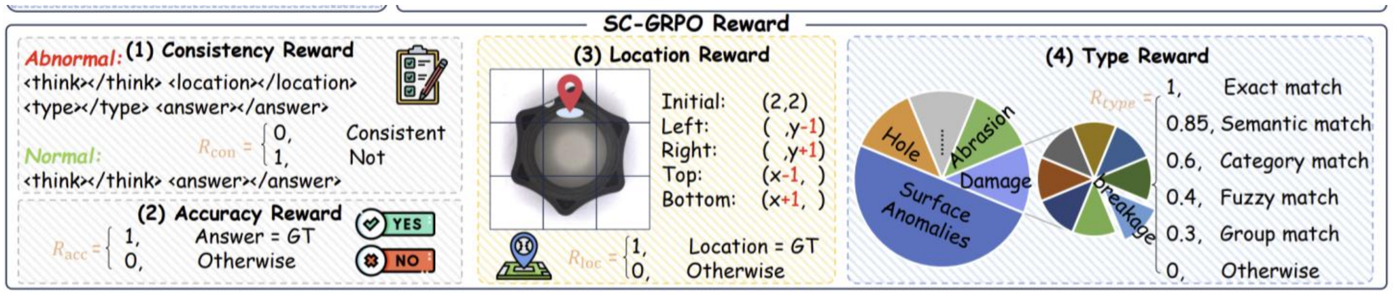
DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models



$$A_i = \frac{R_{\text{SC-GRPO}}(o_i) - \text{mean}\{R_{\text{SC-GRPO}}(o_j)\}_{j=1}^{G}}{\text{std}\{R_{\text{SC-GRPO}}(o_j)\}_{j=1}^{G}}.$$

Methodology

SC-GRPO (Reinforcement Learning Stage)



Multi-dimensional Reward Design

- Consistency (R_con) → reasoning ↔ answer 일관성
- Accuracy (R_acc) → 정답 맞았는지
- Location (R_loc) → 위치 정확한지
- Type (R_type) → 결함 종류 맞는지

$$P_{normal} = \langle think \rangle ... \langle /think \rangle \langle answer \rangle ... \langle /answer \rangle.$$

$$P_{abnormal} = \langle think \rangle ... \langle /think \rangle \langle location \rangle ... \langle /location \rangle \langle type \rangle ... \langle /type \rangle \langle answer \rangle ... \langle /answer \rangle.$$

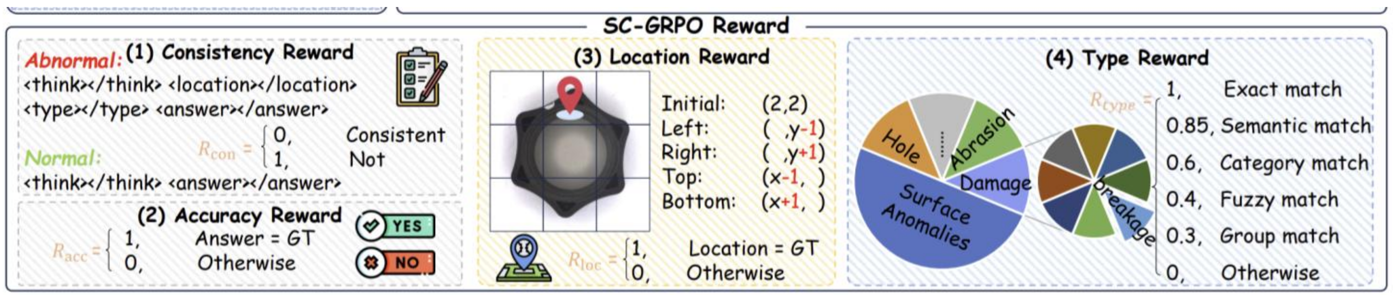
Consistency Reward

- 출력 구조를 강제해서 reasoning 일관성 확보함
- Format 일치하면 1, 불일치하면 0
- Reasoning과 answer간 inconsistency 방지함

$$R_{con}(o_i) = \begin{cases} 1, & \text{if Match}(o_i, P) \\ 0, & \text{otherwise} \end{cases}$$

Methodology

SC-GRPO (Reinforcement Learning Stage)



Accuracy Reward

- Ground-truth랑 직접 비교하여 정확도 평가함
- Correct → 1, Wrong → 0

$$R_{acc}(a_{pred}, a_{gt}) = \begin{cases} 1, & \text{if } a_{pred} = a_{gt} \\ 0, & \text{otherwise} \end{cases}$$

Location Reward

- 이상 위치 정확도 평가
- 텍스트 위치 → 3×3 grid mapping, GT와 비교
- 공간적 이해 학습

$$R_{loc}(l_{pred}, l_{gt}) = \begin{cases} 1, & \text{if } \Phi(l_{pred}) = \Phi(l_{gt}) \\ 0, & \text{otherwise} \end{cases}$$

Type Reward

- 결함 유형 분류 정확도 평가
- 단순 string match가 아닌 다양한 표현 허용
- reward sparsity 완화

$$R_{type}(t_{pred}, t_{gt}) = \begin{cases} 1.0, & \text{if exact match} \\ 0.85, & \text{if semantic match} \\ 0.6, & \text{if category match} \\ 0.4, & \text{if fuzzy match} \\ 0.3, & \text{if group match} \\ 0, & \text{otherwise} \end{cases}$$

ELLab